AIRLOCK – A DRONE TO BASE COMMUNICATION

A PROJECT REPORT

Submitted by

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for the award of the degree

of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING



CHENNAI INSTITUTE OF TECHNOLOGY

(An Autonomous Institution, Affiliated to Anna University, Chennai)



MARCH - 2026

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We jointly declare that the project report on **"AIRLOCK – A DRONE TO BASE COMMUNICATION"** is the result of original work done by us and best of our knowledge, similar work has not been submitted to **"ANNA UNIVERSITY CHENNAI"** for the requirement of Degree of **BACHELOR OF ENGINEERING**. This project report is submitted on the partial fulfilment of the requirement of the award of Degree of **COMPUTER SCIENCE AND ENGINEERING**.

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ABSTRACT

Airlock is a secure, lightweight telemetry system developed for real-time drone communication with improved resistance to interception and replay attacks. The central innovation is in the form of a dynamic encryption system that changes the communication code every two seconds to constantly obfuscate and reduce exposure to static key exploits. The time-based mutation of the code is coordinated across devices to allow effortless decryption while reducing latency. Airlock is designed to be portable in the form of a miniaturized chip-based module that can be integrated into any drone platform. Upon insertion, the chip takes care of telemetry encryption, anti-replay authentication, and USB-based data transfer with ground receivers or external systems. Telemetry logged is retained in a local SQLite database and made available through a RESTful API for integration with dashboards and analytics software. The system supports live visualization of flight metrics altitude, speed, battery status, and trajectory through an interactive web interface. Designed for extensibility and containerized deployment, Airlock offers a scalable solution for secure drone telemetry across simulation and field environments. This research explores the architecture, cryptographic strategy, and performance benchmarks of Airlock, highlighting its potential for next-generation UAV security.

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CHAPTER 1

INTRODUCTION

The growing use of drones across various sectors such as agriculture, defense, logistics, surveillance, and emergency response has created an increasing demand for secure and reliable telemetry systems to ensure safe and uninterrupted operations. Traditional telemetry systems often rely on weak endpoint handling and exposed communication paths, making them highly vulnerable to interception, tampering, and replay-style misuse. These vulnerabilities can compromise mission integrity, leak sensitive operational data, and even lead to loss of control in critical scenarios. To address these challenges, this project proposes Airlock, a secure and portable telemetry framework designed to strengthen drone-to-ground communication through a combination of encryption, validation, monitoring, and modular deployment. Airlock's core innovation lies in its practical three-laptop architecture (Laptop A, Laptop B, and Laptop C), which simulates a complete and secure telemetry pipeline with clearly defined roles for transmission, validation, and visualization. In this setup, Laptop A functions as the telemetry sender by running a Flask-based server that continuously generates and transmits encrypted telemetry packets, including data such as GPS position, altitude, speed, battery level, and flight path. Laptop B acts as the receiver client, where incoming packets are verified for structure, checked for replay-style duplication or abnormal patterns, decrypted, and logged for traceability and auditing. Laptop C hosts a read-only Streamlit dashboard that provides real-time situational awareness by discovering active devices, monitoring packet transmission rates, flagging failsafe conditions, and visualizing live drone movement through an interactive map along with key performance indicators. The system is built around a USB-like secure link simulation, representing a portable, chip-like module that can be integrated into drone systems to enable autonomous encryption handling, anti-replay validation, and reliable device-to-device communication. Telemetry data is persisted in structured local logs (and SQLite-style storage where required) and made accessible through REST API endpoints for seamless integration with dashboards and analytics tools. By combining secure transmission, real-time monitoring, and modular extensibility, Airlock enables efficient visualization of flight statistics while providing a scalable and adaptable solution for next-generation UAV telemetry security in both simulated and practical deployment environments.

1.1 BACKGROUND OF THE STUDY

As drones are increasingly taking on critical roles in fields such as agriculture, defense, logistics, surveillance, and emergency response, there is a growing need for secure and reliable telemetry systems to ensure safe and uninterrupted operations. In many traditional setups, endpoint exposure and unsecured telemetry paths make drone communication vulnerable to interception, tampering, and replay-style misuse, which can disrupt data flow, compromise mission integrity, and pose serious operational risks. Such weaknesses highlight the necessity for a more robust and practical security framework that not only protects data in transit but also verifies its authenticity and ensures continuous monitoring throughout the communication process.

To address these challenges, the Airlock framework proposes a practical and implementation-focused three-laptop architecture using Laptop A, Laptop B, and Laptop C, significantly enhancing the security and observability of the telemetry pipeline. Laptop A functions as the sender, transmitting encrypted telemetry through a Flask-based server, while Laptop B acts as the receiver that validates incoming packets, detects duplicate or abnormal behavior, decrypts valid data, and logs all events for traceability. Laptop C hosts a read-only Streamlit dashboard that enables real-time monitoring through device discovery scanning, packet-rate analysis, failsafe detection, and live map-based visualization of drone movement. Designed as a portable, USB-like secure telemetry pipeline, the system demonstrates how encryption, validation, logging, and visualization can be integrated into a cohesive workflow. This study explores Airlock's architecture and evaluates its performance as a scalable and extensible solution for next-generation UAV communication systems.

1.2 RESEARCH PROBLEM

Classic drone telemetry systems are susceptible to interception, replay attacks, and platform rigidity because of exposed telemetry paths, weak endpoint handling, and limited deployment flexibility across devices. Such flaws represent significant threats in real-time UAV operations, particularly in sensitive or mission-critical-environments. There is an urgent need for a secure, transportable, and implementation-ready telemetry system that can be validated across multiple nodes (Laptop A, Laptop B, and Laptop C) without sacrificing performance or scalability.

1.3 RESEARCH QUESTIONS

1. How effective is the Laptop A–B telemetry encryption and validation flow in inhibiting interception and replay-style misuse in the Airlock drone telemetry system?

2. Is it possible to implement a portable Airlock module reliably across different drone setups using the three-laptop architecture for safe and reproducible telemetry communication?

3. How is USB-like device-to-device telemetry exchange (A → B) under the current Airlock implementation in terms of latency, throughput, and reliability?

4. How does the Airlock encryption, logging, and verification pipeline affect telemetry data integrity and real-time visualization on Laptop C?

5. How scalable is the Airlock framework when utilized across multiple simulated telemetry streams and extended monitoring conditions?

6. What are the performance and usability thresholds of the Airlock Streamlit dashboard in discovering devices, detecting failsafe conditions, and tracking real-time flight data?

7. Is Airlock capable of being a standardized solution for secure UAV telemetry across a range of operational scenarios using the current multi- device implementation?

8. How effectively does the Airlock system maintain secure telemetry continuity under adverse conditions such as packet loss, delayed delivery, network instability, or high traffic bursts, while still preserving validation accuracy and dashboard responsiveness?

9. How does the Airlock system handle scalability when multiple drones simultaneously transmit telemetry to the system without affecting performance or validation accuracy?

10. To what extent can the Airlock framework be adapted from the current laptop-based implementation to real embedded UAV hardware without compromising reliability and efficiency?

1.3.1 RESEARCH OBJECTIVES

The objective of this study is to confirm the validity of the Airlock encryption and validation pipeline in making interception and replay-style misuse impractical across the telemetry communication link. This involves demonstrating smooth decryption, accurate packet verification, and stable synchronization between Laptop A (sender) and Laptop B (receiver) even under continuous and high-frequency telemetry streaming conditions. The evaluation also focuses on how effectively packet identifiers and validation logic can detect duplicates, abnormal patterns, and potential disruptions without affecting normal data flow. In addition, the portability of the Airlock framework is assessed by analyzing whether the current implementation can be adapted into a compact, secure module suitable for integration into different drone platforms while maintaining performance efficiency and minimal computational overhead.

Alongside this, the system is evaluated in terms of real-time throughput, latency, and overall reliability of the USB-like telemetry exchange (A → B), ensuring that encryption, validation, and logging processes do not interfere with live operation or introduce noticeable delays. The usability and scalability of the REST API and Streamlit dashboard on Laptop C are also examined, particularly in how effectively they support real-time visualization, device discovery scanning, packet-rate monitoring, and security status reporting through alerts and failsafe indicators. Furthermore, the robustness of the Airlock system is benchmarked through simulated telemetry runs and practical multi-device testing scenarios, including stress conditions such as high packet rates, duplicate transmissions, and unstable communication behavior, to ensure that the system maintains integrity, responsiveness, and monitoring clarity under both normal and adverse operating environments.

1.3.2 STARTING CODE



1.4 SIGNIFICANCE OF THE STUDY

Growing application of drones in high-stakes industries like defense, agriculture, logistics, and surveillance requires robust and secure telemetry systems. This research fills a critical void in UAV communications with Airlock, a telemetry system that merges encryption, validation, monitoring, and multi-device portability through a practical three-laptop setup. Its novelty is how it implements a secure telemetry pipeline across Laptop A, Laptop B, and Laptop C, significantly lowering the chances of interception, tampering, and replay-style misuse beyond a single-node or unsecured telemetry flow. By structuring the system as a portable, chip-like secure module concept, Airlock provides a plug-and-play approach that can be adapted across drone platforms, while being validated through repeatable device roles in the current implementation. The addition of USB-like device-to-device transfer simulation and real-time visualization tools improves operational transparency and decision-making. This work aids in the creation of next-generation UAV security protocols, presenting a practical, extensible, and high-performance solution for simulated telemetry and real- time monitoring environments. It provides a foundation for future innovation in secure communication of drones.

1.5 SCOPE OF THE STUDY

This work addresses the design, implementation, and testing of Airlock, a secure, portable, and lightweight telemetry system for UAVs. The work covers the design of an encrypted telemetry pipeline using Laptop A, Laptop B, and Laptop C to validate secure transmission, verification, and real-time monitoring in a multi-node environment. It also encompasses a portable chip-like module concept that enables autonomous encryption handling, detection of replay-style misuse through validation logic, and a USB-like device-to-device telemetry exchange between the sender and receiver systems. The approach ensures that telemetry is not only protected during transmission but also verified and logged at the receiver side for integrity and traceability.

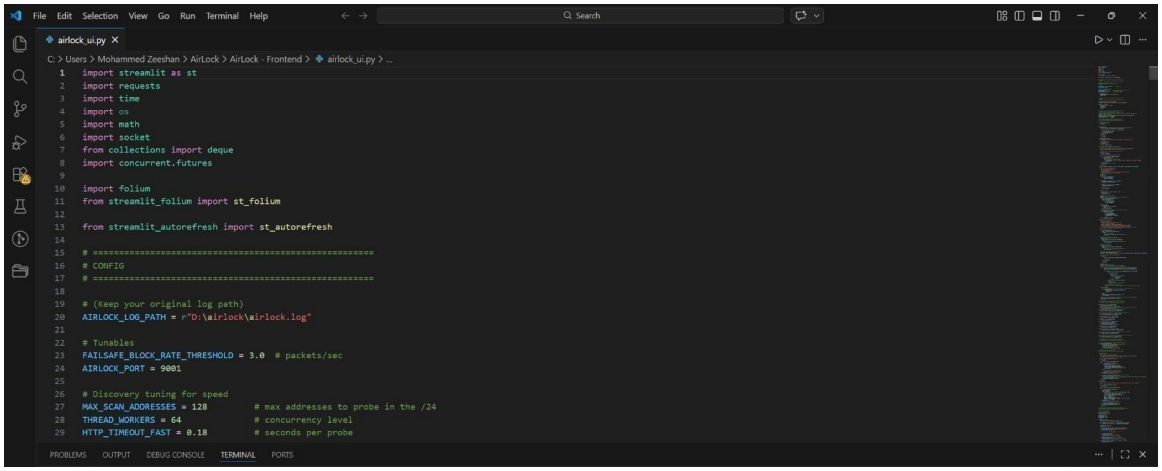
The research further covers real-time telemetry logging through structured local logs and SQLite-style storage, along with the development of REST API endpoints for transmitting, receiving, and monitoring telemetry data. An interactive Streamlit dashboard is implemented to visualize flight metrics such as GPS position, altitude, speed, battery level, and path tracking, while also displaying packet-rate monitoring, anomaly alerts, and system status indicators. Testing is conducted in simulated as well as practical multi-device environments, focusing on performance, validation accuracy, scalability, and system robustness.

1.5.1 CORE FUNCTIONALITIES

The Airlock telemetry framework is designed to provide secure, real-time drone-to- ground communication through an integrated set of core functionalities that combine encryption, validation, logging, and live monitoring. At its core is an encrypted telemetry pipeline implemented across Laptop A, Laptop B, and Laptop C, which significantly reduces the likelihood of interception, tampering, and replay-style misuse during transmission. Laptop A operates as the telemetry sender, transmitting encrypted packets through a Flask-based REST interface, while Laptop B functions as the receiver client that accepts incoming packets, performs structural and behavioral validation checks, decrypts legitimate payloads, and logs the results for traceability. Additional protection is achieved through packet identifiers and rate- based verification logic that can identify duplicated, delayed, malformed, or suspicious telemetry behavior before it affects system integrity. This structured sender–receiver validation flow ensures that telemetry remains both protected in transit and verified upon reception, reinforcing confidentiality and authenticity within the communication channel.

The system is engineered around a portable, chip-like module concept that can be adapted for integration into various drone platforms, while the current implementation validates the design using a USB-like device-to-device exchange between Laptop A and Laptop B. Telemetry data including altitude, speed, battery charge, and positional coordinates is recorded into structured local logs (and SQLite- style storage where required) and made accessible through REST API endpoints that support sending, receiving, monitoring, and dashboard integration. On Laptop C, an interactive Streamlit dashboard provides real-time situational awareness by displaying live flight metrics, packet-rate indicators, security status flags, anomaly alerts, and route tracking on a dynamic map interface. The modular architecture supports extensibility and scalable monitoring, allowing the framework to operate reliably in simulated test environments and practical multi-device conditions, while maintaining a clear separation between transmission, validation, logging, and visualization layers.

1.5.2 UI CODE FORMAT



1.5.3 LIMITATIONS

Although the Airlock framework presents major improvements in secure drone telemetry, there are some limitations revealed in this research. First, the current multi- device telemetry flow albeit efficient depends on stable connectivity and consistent device coordination across Laptop A, Laptop B, and Laptop C, which can prove difficult in environments with unstable networks, changing IPs, or delayed packet delivery. Second, the portable module concept is designed for general drone platforms but could still need hardware-specific adaptation for particular UAV models, restricting universal compatibility without customization-in-real-deployments.

Moreover, the device-to-device telemetry exchange (USB-like link simulation), although effective for short-range and direct links, can prove ineffective for long- range or wireless telemetering operations without an added secure transport layer. Performance of the system has been assessed mainly in simulated telemetry runs and controlled multi-laptop conditions; deployment under real- world harsh environments (e.g., high-interference zones or high-risk operations) is beyond the purview of this work. Lastly, the dashboard and API that are available today are optimized for real- time monitoring, discovery, visualization, and logging but do not contain advanced analytics, anomaly prediction, or automated incident response, which could be considered in subsequent iterations.

Despite its strengths, AirLock has certain limitations, including reliance on simulated multi-device testing rather than full embedded drone hardware deployment, evaluation primarily in single-stream scenarios rather than large multi-drone swarms, and the use of rule-based intrusion detection that may not detect highly sophisticated or adaptive threats.

CHAPTER 2

LITERATURE REVIEW

2.1. INTRODUCTION TO DRONE SECURITY RESEARCH

The explosive growth of unmanned aerial vehicles (UAVs) in the defense, industrial, and civilian markets has created new opportunities in surveillance, logistics, agriculture, and disaster management. But this adoption has also generated some serious cyber security implications. Drones differ from traditional networked systems in that they are mobile, limited in resources, and deployed in hostile or remote environments. This renders them prime targets for cyber threats. GPS spoofing, command injection, firmware tampering, and denial-of-service attacks reports underscore the imperative for strong, multi- layered defence systems.

The community responded by investigating a range of approaches, from hardware- based trust (TPM, PUFs) to light-weight cryptography (ECC, AES), blockchain-based accountability, and intrusion detection systems. This literature survey covers the most pertinent contributions and places them against the background of the AirLock CCP protocol, whose originality is to offer a USB- activated, end-to-end multi-layer security solution for drones.

2.2. TRUSTED COMPUTING AND SECURE BOOT

One of the oldest and most widely discussed methods for guaranteeing device trust is the Trusted Platform Module (TPM), which establishes a hardware-based chain of trust starting from the boot process. TPMs securely store cryptographic keys, verify firmware integrity, and ensure that only authenticated software is executed on a device. By anchoring trust at the hardware level, they provide strong protection against unauthorized access, tampering, and malicious code injection. However, while TPMs offer high security, they often require specialized hardware integration, which can increase cost and reduce flexibility in lightweight or portable systems such as UAV telemetry frameworks. By checking that every level of software from the bootloader all the way up to the operating system has not been modified.

Wang et al. (2017) presented a better embedded trusted computing model that combines Trusted Cryptography Module (eTCM) and Trusted Platform Control Module (eTPCM). Symmetric and asymmetric cryptographic primitives, as well as active defense mechanisms, are employed in their design to enforce a secure boot process. Their system keeps unauthorized code from execution by embedding cryptographic proofs of integrity within FPGA modules.

AirLock extends this concept further by implementing secure boot through its USB utility, which initiates cryptographic verification before allowing the drone firmware to launch. As opposed to fixed, permanently integrated TPMs, AirLock supports modular activation and is therefore highly appropriate in fleet environments where operators need flexibility in control. USB-based architecture is also easier to service and update, as the security code is in an external utility.

Moreover, secure boot research points to threats such as rollback attacks, wherein attackers re-install vulnerable versions of firmware. AirLock addresses this by employing session-based keys associated with validated firmware, such that only the latest and approved versions of firmware can initiate communication channels.

2.3. PHYSICALLY UNCLONABLE FUNCTIONS (PUFs)

While TPMs depend on embedded hardware modules, a second research thrust investigates Physically Unclonable Functions (PUFs) as minimalist authentication primitives. PUFs use the uncontrollable variability of semiconductor manufacturing to generate unclonable hardware signatures.

Pal et al. (2020) proposed a PUF-based drone security framework where the flight controller, companion computer, and sensors are tied to their individual PUF responses. This keeps counterfeit parts or cloned drones out of a network. Since the PUF response is not stored but rather on-demand generated, attackers cannot just obtain cryptographic keys by extracting them.

The PUF method is very efficient but has its limitations—it is based on specialized FPGA hardware, which makes it not quite suitable for off-the-shelf drones. Compared to that, AirLock seeks similar hardware trust levels without requiring redesign of the hardware, based on a USB dongle as a trust anchor that is portable. With this, AirLock is very deployable while still enjoying the advantages of dynamic key generation and authentication at the hardware level.

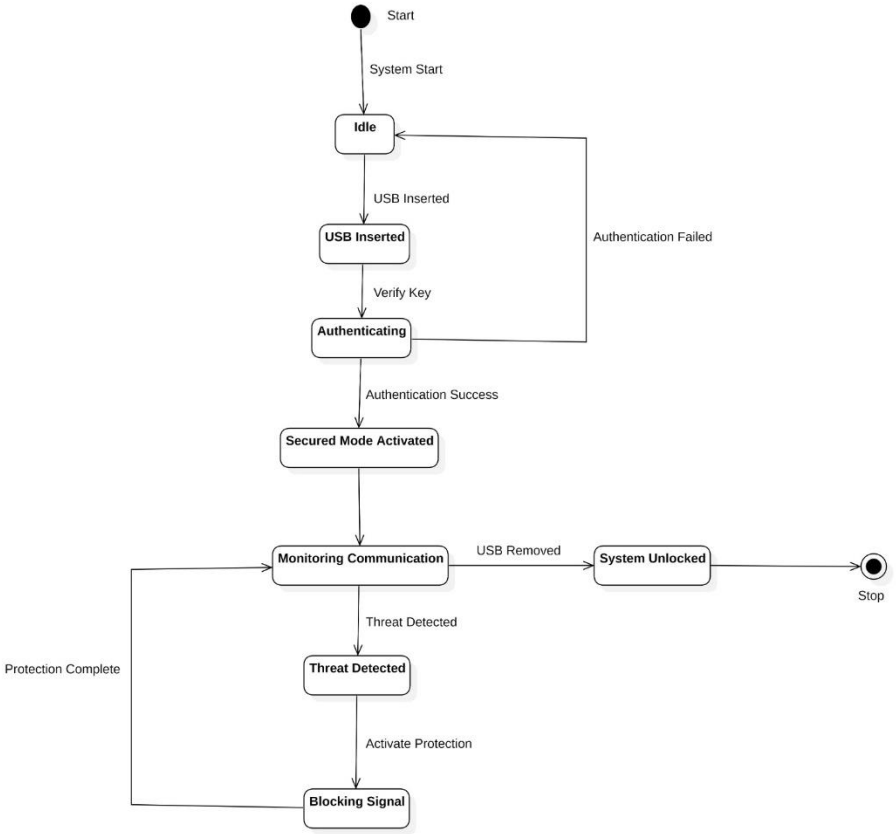
2.3.1. CRYPTOGRAPHY FOR RESOURCE-CONSTRAINED UAVS

Drones typically operate with limited processing power, memory, and battery capacity. Hence, heavyweight cryptographic algorithms like RSA are often impractical. Research has therefore focused on lightweight cryptography, particularly Elliptic Curve Cryptography (ECC).

ECC provides the same level of security as RSA but with much shorter key sizes. Liu and Zhou (2019) illustrated that ECC can be well-fitted for low-power devices of the IoT with effective encryption, decryption, and key exchange without draining resources. Likewise, El-Zawawi et al. (2022) proposed BDIVE, a blockchain-supported drone authentication scheme using ECC and distributed ledgers. Their assessment indicated dramatic declines in energy consumption (70%) and computational overhead (68%) over RSA-based protocols.

AirLock incorporates ECC in its exchange of key phase. On the insertion of the USB dongle, it creates a new ECC key pair and does Elliptic Curve Diffie– Hellman exchange with the drone. The common secret is hashed into a symmetric AES key employed for the encrypted session. Forward secrecy is guaranteed, such that even if one session key were compromised, previous communications are secure. The pairing of ECC for key exchange and AES for data encryption is a compromise between efficiency and strength.

2.4. FLOW DIAGRAM



2.5. BLOCKCHAIN AND IMMUTABLE LOGGING

Yet another critical research area in UAV security is auditability and accountability. Blockchain and associated technologies are increasingly utilized to provide tamper- proof logging of drone operations.

Chaining logs with cryptographic hashes has been demonstrated by researchers to make altering previous records essentially impossible to detect. El-Zawawi et al. (2022) incorporated blockchain in their ECC-based authentication system to enable traceability in drone-assisted Internet of Vehicles (IoV) scenarios. These logging mechanisms are essential in applications involving the use of drones for surveillance, law enforcement, or military operations, where forensics after the operation may be necessary. AirLock applies this notion in its Secure Logging and Audit Layer. Rather than utilizing full blockchain implementation, which is computationally expensive, AirLock applies hash chaining to generate immutable logs. Every log entry points to the previous one, creating a chain that safeguarded against replay attacks as well as silent tampering. It has most of the advantages of blockchain with much lower resource overheads, which makes it appropriate for UAV use.

2.5.1. USB-BASED AUTHENTICATION AND HARDWARE TOKENS

There is a distinct body of research that focuses on token-based physical security, specifically USB dongles, as the means to implement strong authentication. Clarke (2018) pointed out the benefits of USB tokens in IoT environments, that they are immune to attacks at a distance because the attackers have to hold the token physically.

Following this concept, some researchers have tried embedding cryptographic primitives within USB modules to improve endpoint security. These implementations safeguard against man-in-the-middle attacks by ensuring the performance of cryptographic operations within the dongle.

AirLock's innovation is to take this process beyond authentication. Not only does its USB dongle authenticate but also initiate secure boot, start ECC key exchange, initiate encrypted communication, and impose audit logging. This integrates the USB module as an active member of the drone's security lifecycle, rather than merely a passive carrier of credentials.

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2.6. UAV INTRUSION DETECTION SYSTEMS

Encryption itself cannot provide absolute security; real-time monitoring is required to identify anomalies. Intrusion Detection System (IDS) research in UAVs aims at detecting abnormal traffic patterns, abnormal timing behaviors, and unauthorized command injections.

Kumar and Singh (2021) have employed machine learning algorithms on UAV traffic data and proved successful in detecting replay and jamming attacks. IDS solutions for drones tend to use lightweight anomaly detection algorithms to prevent overloading of resource-starved platforms.

AirLock fills the gap by implementing an Intrusion Detection Layer that relies on SHA- 256 with message IDs to ensure message integrity and identify tampering. If issues like packet reordering or duplication are identified, AirLock initiates an emergency lockdown procedure, terminating communication at once to avoid further exploitation. This aligns with the general IDS literature while providing a context-specific adaptation for drones.

2.7. POST-QUANTUM CRYPTOGRAPHY (PQC)

A growing future concern in drone security is the emergence of quantum computing, which has the potential to weaken widely used public-key algorithms such as ECC and RSA that currently protect many communication systems. As quantum capabilities advance, algorithms based on classical mathematical hardness assumptions may become vulnerable to large-scale quantum attacks. Researchers such as Chen et al. (2022) emphasize the urgent need to design and adopt post-quantum cryptographic algorithms that can resist quantum-enabled adversaries while remaining efficient enough for constrained platforms like UAVs. For frameworks like AirLock.

While PQC is not yet in general use in UAVs, it is an interesting aspect of future- proofing security measures. AirLock has been built as a cryptographically flexible model, which implies that it is capable of incorporating PQC schemes once they are lightweight and standardized. This makes AirLock not only efficient in existing environments but also capable of handling future threats.

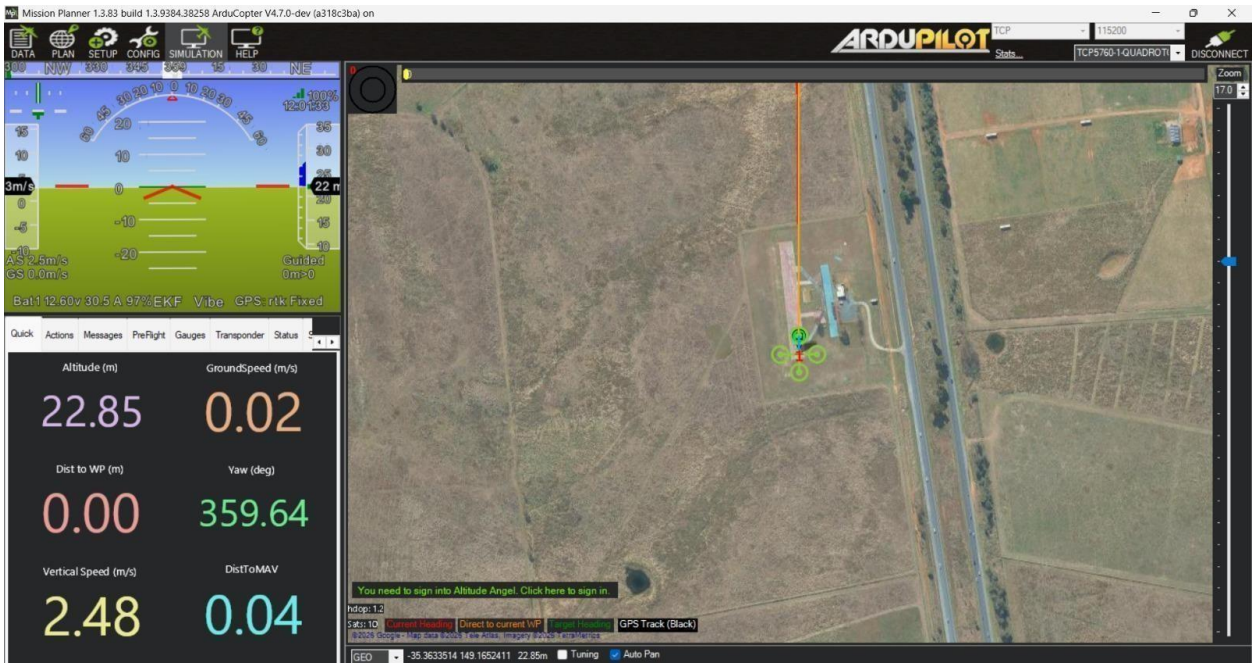
2.8. COMPARATIVE INSIGHTS

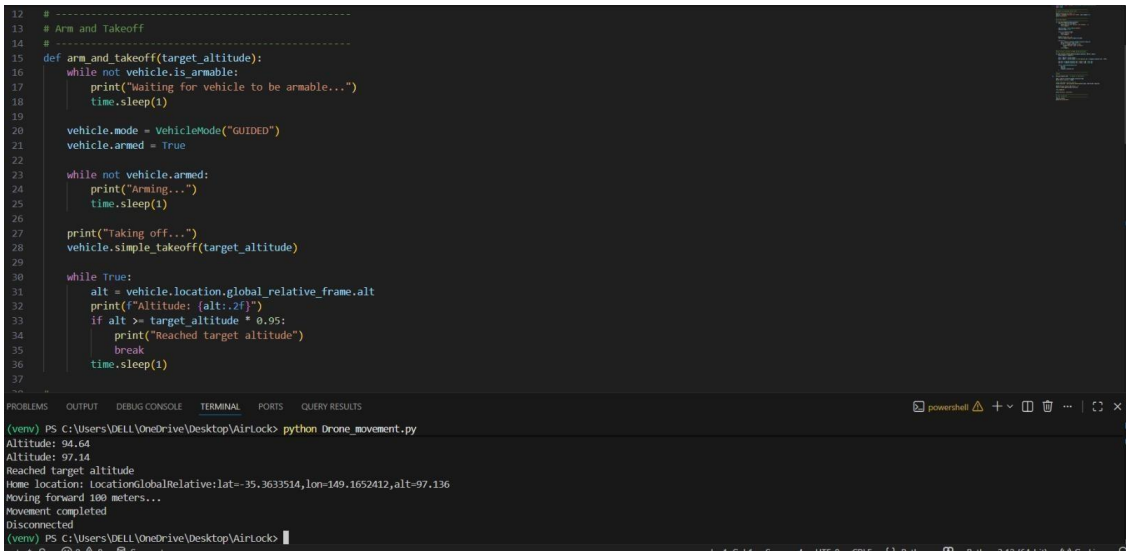
The literature discussed hereunder identifies numerous approaches that align with the current AirLock implementation and its future upgrades. Several works highlight hardware-rooted trust concepts, where identity and authorization are tied to a dedicated module; while these methods are robust, they may require specialized embedded components and UAV-specific adaptation. Research also consistently supports lightweight encryption and verification techniques for real- time telemetry, since they can secure data streams with low overhead. Integrity- focused mechanisms such as tamper-evident logging and chained records are emphasized for accountability and auditing, helping verify whether telemetry has been altered or replayed.

In addition, portable device-to-device security designs (token-like modules or dongle- style secure links) are presented as practical and deployable for controlled short-range exchange, which connects to AirLock's USB-like telemetry transfer and multi-device validation flow. IDS-style monitoring is also identified as essential for resilient UAV operations, enabling real-time detection of abnormal packet behavior, suspicious rates, and disruption patterns. Finally, modern literature points toward API-driven architectures and dashboards as critical for real-world usability, since live visualization improves operational clarity, while forward-looking work in post-quantum cryptography continues to guide how systems like AirLock can stay secure against emerging threats.

AirLock is unique in combining these threads into a unified USB-initiated protocol. In contrast to previous studies, which tend to emphasize a single layer of protection, AirLock offers a layered defense: from boot integrity to encrypted communication, intrusion detection, and immutable logging. This integrated strategy situates AirLock as a deployable, scalable, and future-proof drone cybersecurity framework.

2.8.1. DRONE SIMULATION





CHAPTER 3

METHODOLOGY

3.1 RESEARCH DESIGN

The research approach of the AirLock project is founded on a design science research approach that focuses on problem articulation, system design, implementation, and evaluation. Because drones are susceptible to cyber threats like spoofing, replay attacks, and unauthorized firmware tampering, our design centers on the incorporation of cryptographic security features with hardware- accommodated trust mechanisms.

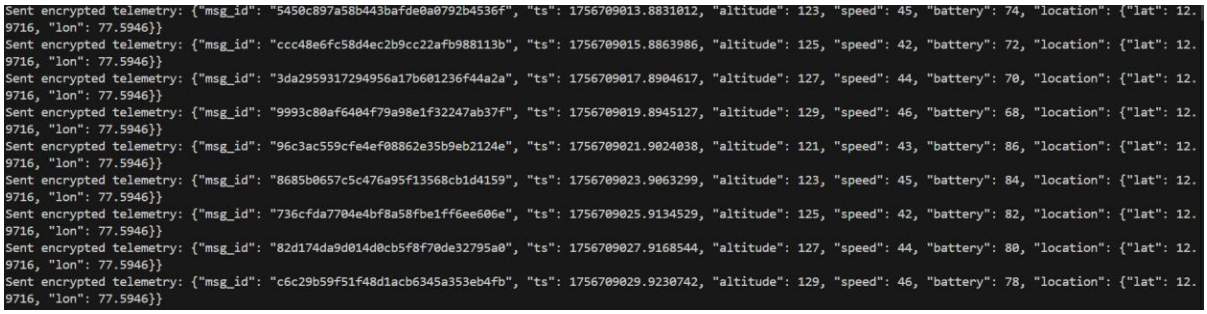
The research approach adheres to three phases: an exploratory phase that examines current drone telemetry security architectures, identifies UAV communication vulnerabilities, and reviews practical encryption, validation, and monitoring methods suitable for real-time telemetry; a development phase that builds the AirLock proof of concept in Python using the current multi-device setup, where Laptop A transmits encrypted telemetry through a Flask server, Laptop B receives, decrypts, validates, and logs packets, and Laptop C runs a Streamlit dashboard for discovery scanning, failsafe detection, and live map-based visualization; and an evaluation phase that emulates drone-to-ground communication across the A → B pipeline, executes disruption and replay-style test cases (such as duplicated packets and abnormal rate conditions), and measures system reliability, dashboard responsiveness, and end-to-end telemetry integrity. This structured approach ensures the work remains both rigorous and practical, contributing to scholarly understanding while delivering a deployable telemetry security workflow.

3.2 DATA COLLECTION METHODS

Secondary Secondary data collection and simulation data generation are used in the project, combining research-backed evidence with practical experimentation aligned to the current AirLock implementation. Academic literature is reviewed from sources such as IEEE, Springer, and the ACM Digital Library, focusing on UAV telemetry security, encryption and integrity validation methods, anti- replay concepts, secure communication pipelines, and real-time monitoring approaches that can be applied without heavy computational overhead. Standards and protocol references are also consulted to guide secure design choices, including widely accepted cryptographic recommendations

For experimentation, simulation data is generated as synthetic telemetry streams such as GPS coordinates, altitude, speed, battery level, and sensor- style readings produced through Python-based telemetry generators and simulator logic. These telemetry packets are then transmitted through the implemented A → B pipeline, where Laptop A sends encrypted telemetry via a Flask server, Laptop B receives, validates, decrypts, and logs the packets, and Laptop C visualizes the live status through the Streamlit dashboard with discovery scanning, rate monitoring, failsafe alerting, and map-based tracking. To evaluate robustness, disruption and attack-style scenarios are introduced, including replay-style duplication of packets, injection-like abnormal transmissions, delayed delivery, and corruption-style manipulation attempts, allowing the system's validation checks, logging traceability, and monitoring alerts to be assessed under stream

3.3 SYSTEM DESIGN & ARCHITECTURE



3.3.1 TOOLS AND TECHNOLOGIES USED

The project is implemented using Python 3.12, chosen for fast prototyping and smooth development of multiple components in a single ecosystem. Flask is used to host the REST-based telemetry sender on Laptop A, enabling structured communication to Laptop B, which runs the receiver client responsible for accepting packets, validating them, decrypting them, and logging the results. For communication and testing, HTTP request utilities are used to transmit telemetry and verify send/receive behavior, while socket and network utilities support device- to-device simulation and local discovery- style scanning used by the monitoring layer. Telemetry and security events are persisted through local log storage (and SQLite-style structured storage where required) to maintain traceability of packet flow, validations, and failsafe triggers. For visualization, the system uses a Streamlit dashboard on Laptop C, enhanced with Folium/streamlit-folium for map-based tracking and live monitoring, allowing real- time display of telemetry metrics, security indicators, device discovery results, and operational status.

3.3.2 MATERIALSAND DATASETS

The datasets used in this project consist primarily of synthetic telemetry logs generated through the telemetry simulator during continuous streaming runs, including parameters such as GPS coordinates, altitude, speed, battery levels, and flight-path points. Communication datasets are captured in the form of encrypted telemetry packets exchanged between Laptop A and Laptop B, along with receiver-side validation traces that determine packet legitimacy and flag duplicates or abnormal behavior. Audit-related data is maintained through structured local logs (and SQLite-style storage where applicable), preserving a complete history of transmissions, validation outcomes, and monitoring events for traceability. Additionally, attack datasets are created by introducing replay-style duplicated packets, abnormal-rate bursts, delayed transmissions, and corruption-like manipulations, which are used to test the system's failsafe mechanisms, validation accuracy, and the responsiveness of the monitoring dashboard.

3.3.3 PROCEDURES

The current AirLock workflow begins by assigning clear roles across the three- device setup, where Laptop A acts as the telemetry sender, Laptop B operates as the receiver and validator, and Laptop C runs the read-only monitoring dashboard. Telemetry is then generated using the simulator logic to produce continuous packets containing values such as GPS position, altitude, speed, battery level, and flight-path coordinates. Laptop A encrypts and transmits these packets through the Flask REST API, after which Laptop B receives the data, performs verification checks to identify duplicates or suspicious patterns, decrypts only valid payloads, and logs both the telemetry and validation outcomes for traceability. System integrity is supported by maintaining a consistent record of packet flow and receiver decisions, enabling detection of anomalies such as abnormal packet rates, repeated transmissions, and disruption-style behavior that can trigger failsafe-style flags. Finally, Laptop C continuously discovers active nodes and visualizes live flight and security status through the Streamlit dashboard, showing real-time indicators and map-based tracking to support operational monitoring.

3.4. DATA FLOW AND ENCRYPTION METHOD

Data flow undergoes seven phases: the telemetry simulator continuously produces drone flight data, after which the AirLock pipeline is initialized across Laptop A, Laptop B, and Laptop C to establish a secure sender–receiver–monitoring structure. Laptop A then prepares each telemetry packet for transmission by applying the project's encryption step before sending it through the Flask REST interface. Laptop B receives incoming packets and performs verification checks to detect duplicates,

abnormal patterns, or disruption-style behavior, and only then processes valid telemetry while rejecting suspicious traffic. Packet integrity is preserved by maintaining receiver-side validation traces and consistent logging of transmission outcomes. Intrusion-style monitoring is supported through rate observation and anomaly flags that can trigger failsafe-style alerts when the flow becomes suspicious or unstable. Finally, all telemetry events, receiver decisions, and monitoring indicators are stored safely in local structured logs (and SQLite-style storage where required) to support auditing and later analysis.

3.4.1 ENCRYPTION WORKFLOW:

The current AirLock implementation strengthens telemetry security through a combination of encryption and intelligent validation across devices. Telemetry packets are encrypted on Laptop A before transmission, ensuring that any intercepted data remains unreadable and protected from unauthorized access. This establishes a strong layer of confidentiality, particularly important in networks where data may be exposed during transit.

On the receiving side, Laptop B performs structured validation of each incoming packet using unique identifiers and verification traces. These elements help confirm the authenticity of the data and ensure that it has not been altered during transmission. By comparing identifiers and tracking packet flow, the system can effectively detect duplicated or irregular packets, allowing it to reject suspicious data before it affects the system.

This mechanism also helps mitigate replay-style attacks, where previously captured packets are resent to disrupt or manipulate the system. Since each packet is checked for uniqueness and consistency, repeated transmissions can be identified and discarded. Additionally, the validation process supports traceability, making it easier to monitor data flow and identify anomalies in real time.

Overall, the AirLock approach delivers a reliable balance of confidentiality, integrity, and operational efficiency. It secures telemetry data while ensuring that communication across the multi-device pipeline remains fast, accurate, and resilient against common network threats.

3.5. ALGORITHM & PSEUDO CODE

Step 1: Assign roles: Laptop A = Sender, Laptop B = Receiver, Laptop C = Dashboard.

Step 2: Start the telemetry simulator to generate live drone data on Laptop A (or feed into A).

Step 3: Run the Flask server/API on Laptop A to expose the telemetry send endpoint(s).

Step 4: Initialize sender-side settings (packet schema, encryption mode, ID rules, rate). Step 5: Generate a telemetry packet (GPS, altitude, speed, battery, timestamp/path).

Step 6: Create a unique message/packet ID and attach sequence/timestamp metadata. Step 7: Encrypt the telemetry payload on Laptop A using the active session settings. Step 8: Transmit the encrypted packet from Laptop A to Laptop B via REST request. Step 9: Receive the packet on Laptop B and validate required fields + packet structure. Step 10: Check duplicate/replay using message ID and a recent-packet cache/queue. Step 11: Validate packet behavior (freshness, ordering, abnormal rate, malformed content).

Step 12: Decrypt the payload on Laptop B and extract telemetry fields.

Step 13: Log the packet result (accepted/rejected), telemetry values, and security status locally.

Step 14: Update Laptop C dashboard (discovery status, packet rate, alerts, and live map tracking).

Step 15: Update internal counters on Laptop B (total packets, valid, duplicate, rejected, anomaly count).

Step 16: Push processed telemetry data to the REST API endpoint for dashboard consumption.

Step 17: Laptop C fetches latest telemetry data through the dashboard/data endpoint.

Step 18: Render real-time KPIs on Laptop C (packet rate, device status, connection health).

Step 19: Plot drone position and movement path dynamically on the live map interface.

Step 20: Display alert messages on the dashboard for duplicates, abnormal rates, or validation failures.

3.5.1. PSEUDO CODE EXAMPLE

```
def airlock_protocol():
    # Step 1: Secure Initialization / Boot Gate if not
    verify_firmware():
        raise Exception("Firmware verification failed")

    # Step 2: Key / Session Setup (ECC-style exchange) drone_priv,
    drone_pub = generate_ecc_keys() base_pub =
    request_base_public_key(drone_pub) shared_secret =
    ecc_exchange(drone_priv, base_pub)

    # Step 3: Session Key Derivation (AES key from shared secret) session_key
    = derive_aes_key(shared_secret)

    # Step 4: Secure Telemetry Messaging Loop for
    msg in telemetry_stream():
        encrypted = aes_encrypt(msg, session_key)
        hash_val = sha256_with_id(encrypted) # includes message_id internally send_to_base(encrypted,

        hash_val)

        # Step 5: Intrusion / Anomaly Checks
        if detect_intrusion(encrypted, hash_val):
            trigger_emergency_shutdown()
            break

        # Step 6: Logging / Audit
        log_message(encrypted, hash_val)
```

3.6. ETHICAL CONSIDERATIONS

AirLock's ethical aspect is centred around responsible innovation by ensuring that stronger telemetry security does not come at the cost of privacy, misuse, or lack of accountability. The system is built to protect sensitive operational telemetry such as location, flight path, and mission status by securing the sender-to-receiver pipeline so that exposed telemetry paths are minimized and unauthorized parties cannot easily observe or tamper with flight data. At the same time, the implementation is designed with integrity of use in mind: AirLock is developed as a defensive security framework focused on protecting legitimate UAV operations, improving resilience against interception, tampering, and replay-style misuse, rather than enabling offensive capabilities or facilitating harmful activities. Transparency and accountability are strengthened through structured logging of telemetry events, receiver validation outcomes, and monitoring alerts, which creates an auditable trail that helps operators and evaluators trace what happened during a session and identify abnormal behavior without hiding failures or silently discarding critical incidents. The monitoring layer is intentionally read-only and oriented toward operational safety, using the dashboard on Laptop C to visualize flight metrics, discovery status, and failsafe-style indicators so that security decisions can be made with clarity and oversight rather than opaque automation. In terms of compliance and responsible deployment, AirLock is designed to align with general secure engineering practices such as minimizing exposed interfaces, restricting access to telemetry endpoints, preserving data only as needed for verification and audit, and avoiding unnecessary collection of personal or sensitive information so that the system can be adapted to privacy-conscious environments and organizational policies. Finally, non-malicious deployment is treated as a key concern: by emphasizing controlled access, receiver-side validation, and defensive monitoring, the project reduces the risk that the same tooling could be repurposed for harmful interception or hijacking.

CHAPTER 4

IMPLEMENTATION

4.1 PROJECT WORKFLOW

The AirLock framework implementation uses a modular workflow so that all subsystems (sender, receiver validation, dashboard, and logging) can be developed and tested separately before full integration. The workflow begins with initialization, where roles are assigned across Laptop A (telemetry sender), Laptop B (receiver/validator), and Laptop C (dashboard), ensuring the pipeline is ready. Next, telemetry generation produces simulated drone packets with message IDs and timing fields to support duplicate detection. In secure communication, Laptop A encrypts and sends telemetry through the Flask REST interface to Laptop B. The receiver module then validates structure, detects replay-style duplicates and abnormal traffic patterns, decrypts valid payloads, and logs outcomes for traceability. Finally, Laptop C provides real-time visualization and monitoring through discovery scanning, security indicators, failsafe alerts, and map-based tracking, while structured logs preserve session events and anomalies for auditing and later analysis.

4.1.1 WORKFLOW



4.2. API DESIGN AND ENDPOINTS

Flask-based REST API acts as the middleware between the telemetry sender (Laptop A) and the receiver/validator (Laptop B), while also supporting the read-only monitoring layer on Laptop C. In the current AirLock implementation, the API provides a structured and repeatable interface for pushing encrypted telemetry packets, receiving receiver-side validation outcomes, and exposing processed telemetry and security signals for real-time visualization. This separation is important because it prevents the dashboard from interacting directly with raw, unverified telemetry, and instead ensures that only validated, “clean” data is surfaced for monitoring and decision-making. It also makes the overall architecture modular, so each component (sender, receiver, dashboard) can be tested independently while still operating through the same communication contract.

Typical endpoints include a send/telemetry route where Laptop A submits an encrypted telemetry payload along with packet identifiers and timing metadata, and Laptop B responds with an acceptance status based on verification checks such as packet structure validity, duplicate/replay-style detection. A receive/status route enables retrieval of the most recent receiver outcomes, counters, and processing traces so that monitoring tools can understand not only what telemetry was received, but how it was handled. For visualization, a dashboard/data endpoint returns structured telemetry in JSON format coordinates, altitude, speed, battery level, timestamps, and any derived fields, allowing Laptop C to render live KPIs, trend views, and map tracking without direct access.

4.3.TELEMETRY HANDLING AND DATA ENCRYPT

Step 1: Drone computes telemetry (latitude, longitude, altitude, speed).

Step 2: Converted to JSON format.

Step 3: AES-GCM encrypted using session key derived from ECC. Step 4:

SHA-256 hash + message ID added for integrity.

4.3.1 DECRYPTION PROCESS

Step 1: Receiver retrieves encrypted data through API. Step 2:

AES-GCM decrypted with session key.

Step 3: Integrity checked using SHA-256 hash. Step 4:

Data sent to visualization dashboard.

This keeps the telemetry data confidential, tamper-free, and authenticated.

4.4. DASHBOARD DESIGN

The dashboard module permits real-time monitoring of drones. It is created with the purpose of offering clarity, usability, and anomaly alert



4.4.1 COMPONENTS OF DASHBOARD

The AirLock dashboard on Laptop C combines visual and status components to provide continuous situational awareness during telemetry streaming. It includes a live map view that displays the drone's real-time GPS position and route using the Streamlit map layer (Folium/Leaflet-based), allowing operators to track movement as packets arrive. Alongside this, the interface presents key metric indicators such as altitude, speed, battery level, packet rate, and receiver health as quick-read KPIs, making it easy to understand flight and communication status at a glance. Trend charts are used to show how values like altitude and speed change over time, helping identify unstable patterns or sudden shifts in behavior. A dedicated alert panel highlights security and stability issues such as repeated or duplicate packets, abnormal packet-rate conditions, receiver validation failures, and failsafe-style triggers, ensuring potential disruption is visible immediately. Finally, a log viewer surfaces the most recent telemetry and security events (for example the latest 20 entries) retrieved from local logs or SQLite- style storage, supporting quick auditing and traceability. Through the integration of graphical and textual components, the dashboard grants the operators situational awareness.

4.5. SYSTEM INTEGRATION WITH RECEIVER CLIENT

System connects drone → base → dashboard subsystems via the following modules: Laptop A runs the Drone Simulator, a Python script that generates continuous telemetry packets (GPS, altitude, speed, battery, path) and forwards them as encrypted payloads to the REST endpoint. Laptop B runs the Receiver Client, which accepts the incoming packets through the Flask API, performs validation checks (such as duplicate detection and abnormal-rate filtering), decrypts valid payloads, and logs both telemetry and security outcomes for traceability. An Intrusion-Style Monitoring Module operates through receiver-side verification and rate-based .

4.6. ECC KEY GENERATION & AES SESSION KEY

ECC key generation and AES session key derivation form the foundation of secure session establishment within the AirLock communication workflow. In a typical secure exchange model, elliptic curve cryptography is used to generate lightweight yet strong public–private key pairs for both communicating entities, enabling them to derive a shared secret without directly transmitting it over the network. This shared secret is then processed through a key derivation function to produce a symmetric session key, which is used for efficient high-speed encryption of telemetry data. The advantage of ECC lies in its ability to provide strong cryptographic security with smaller key sizes

compared to traditional algorithms like RSA, making it especially suitable for resource-constrained platforms such as UAVs where processing power, memory, and energy are limited. Once the shared secret is established, an AES-based session key enables fast and efficient symmetric encryption for continuous telemetry streaming, ensuring confidentiality of GPS coordinates, altitude, speed, battery levels, and other mission-critical data. This combination of lightweight key exchange and high-speed encryption allows the system to maintain secure communication without introducing significant latency, thereby supporting real-time operations. Additionally, the use of session-based keys enhances security by limiting the exposure window of any single key, reducing the impact of potential compromise and strengthening overall communication resilience.

4.6.1 SESSION KEY REPRESENTATION

#	msg_id	ts	alt	spd	bat	lat	lon
1	193288fc56574577b23f71a336a7c0f8	1756706861.716	121	43	86	12.9716	77.5946
2	afb20efb9d3a4d829476c21c701a374f	1756706859.713	129	46	68	12.9716	77.5946
3	e2a14e625ca742a09b033052a25feaec	1756706857.711	127	44	70	12.9716	77.5946
4	1692bd8e6f7f4383855db04106f3005f	1756706855.709	125	42	72	12.9716	77.5946
5	0417da9a7ceb4b6cadebbbb83ee25f2c	1756706853.706	123	45	74	12.9716	77.5946
6	1401828ea98a4ffcb71fe3fbfe992d63	1756706851.700	121	43	76	12.9716	77.5946
7	dc4eebfa3c4c40b5a51dfce475b6ac3c	1756706849.697	129	46	78	12.9716	77.5946
8	02f591a0d5c3418abe2ddb36345c48aa	1756706847.694	127	44	80	12.9716	77.5946
9	52d24aac0e014fbab0a62ec82eee7d89	1756706845.689	125	42	82	12.9716	77.5946
10	b6a31b28823040cd8d2d39b841c36718	1756706843.683	123	45	84	12.9716	77.5946
11	54bd04452f75472685aa5b4d60baf6b4	1756706841.681	121	43	86	12.9716	77.5946
12	27eb6f990cdc44a1bec1a99cea9036bb	1756706839.678	129	46	68	12.9716	77.5946
13	927401f6f48d45569e95530cee835026	1756706837.674	127	44	70	12.9716	77.5946
14	073ee5f763974c4f8a1f88a4acc0c685	1756706835.671	125	42	72	12.9716	77.5946
15	af138fb5067f4450b43fe94ae6104cda	1756706833.669	123	45	74	12.9716	77.5946
16	03ef609e4fd34a0bb09aeff9c8895a12	1756706831.668	121	43	76	12.9716	77.5946
17	d40a6e6ab09e4cc0af0e04e0f5f4938a	1756706829.660	129	46	78	12.9716	77.5946
18	ec14e8592c544d8583f7d03aa85dfd07	1756706827.651	127	44	80	12.9716	77.5946
19	740dfeb3d9b94cbeb4537a080a3f3d68	1756706825.649	125	42	82	12.9716	77.5946
20	99fdac406bc84fd68841f217d9b6b137	1756706823.646	123	45	84	12.9716	77.5946
21	5a7a783c9dfd4deabe6eca303a3c21df	1756706821.643	121	43	86	12.9716	77.5946

4.7. COMPARISON BETWEEN UAV TELEMETRY SYSTEMS AND AIRLOCK FRAMEWORK

Feature	Traditional UAV Telemetry	AirLock (Current Implementation)
Communication Model	Direct sender → receiver	Multi-node (Laptop A → B → C) pipeline
Security Approach	Minimal / basic encryption	Encrypted + validated telemetry pipeline
Replay Protection	Limited or absent	Packet ID + duplicate detection
Data Integrity	Not consistently verified	Receiver-side validation + checks
Monitoring	Basic or none	Real-time Streamlit dashboard (Laptop C)
ssVisualization	External tools required	Built-in live map + KPIs + alerts
Logging	Basic logs	Structured logs + SQLite-style storage
Intrusion Detection	Rarely implemented	Rule-based anomaly detection (rate, duplicates)
Scalability	Limited	Modular, extendable to multi-device
Deployment	Fixed hardware setups	Portable, USB-like module concept
Real-Time Performance	Depends on setup	Optimized for continuous streaming
Observability	Low	High (alerts, counters, status flags)
Flexibility	Rigid	Modular and extensible architecture

CHAPTER 5

RESULTSAND FINDINGS

5.1 PRESENTATION OF RESULTS

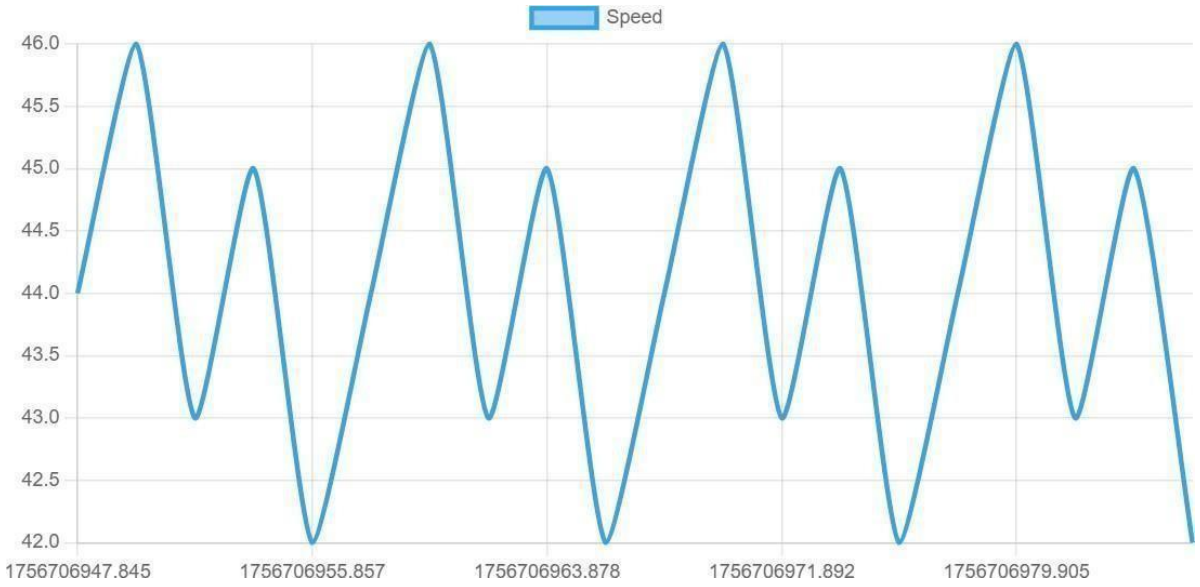
The AirLock prototype was deployed and tested using a Python-based drone simulator integrated with a Flask-based REST API to simulate real-time telemetry exchange. The USB-based initialization mechanism effectively validated the boot sequence, ensuring that only authorized sessions were allowed to proceed, while also supporting the generation of ECC key pairs and the derivation of AES session keys for secure communication. This setup enabled encrypted telemetry transmission along with receiver-side validation, logging, and monitoring, demonstrating that secure session establishment and protected data exchange can be achieved efficiently within a controlled multi-device environment.

5.2. KEY OUTCOMES

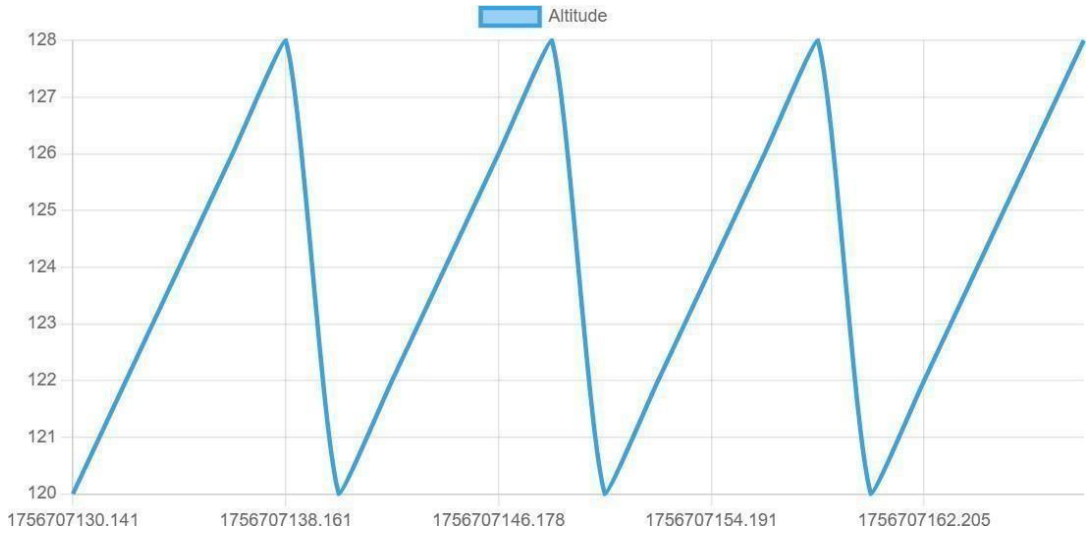
The results show that AirLock achieved a secure and observable telemetry workflow through the three-laptop implementation. The pipeline was reliably initialized with Laptop A as the sender, Laptop B as the receiver/validator, and Laptop C as the monitoring dashboard. Telemetry data such as GPS, altitude, speed, battery, and route points was encrypted on Laptop A and transmitted through the Flask REST interface, while Laptop B verified packets before decrypting and logging them. Integrity was supported through packet IDs and receiver-side validation traces, enabling detection of duplicates, delays, and abnormal packet-rate behavior. When replay-style duplicates and repeated IDs were introduced, the system flagged and rejected them, and the dashboard displayed alerts and failsafe-style indicators in real time. Structured local logs (and SQLite-style storage where required) preserved telemetry history and anomaly events for auditing. These results validate that AirLock achieves confidentiality, integrity, and practical verification over the UAV– ground telemetry communication link.

5.2. SYSTEM WORKFLOW DIAGRAM

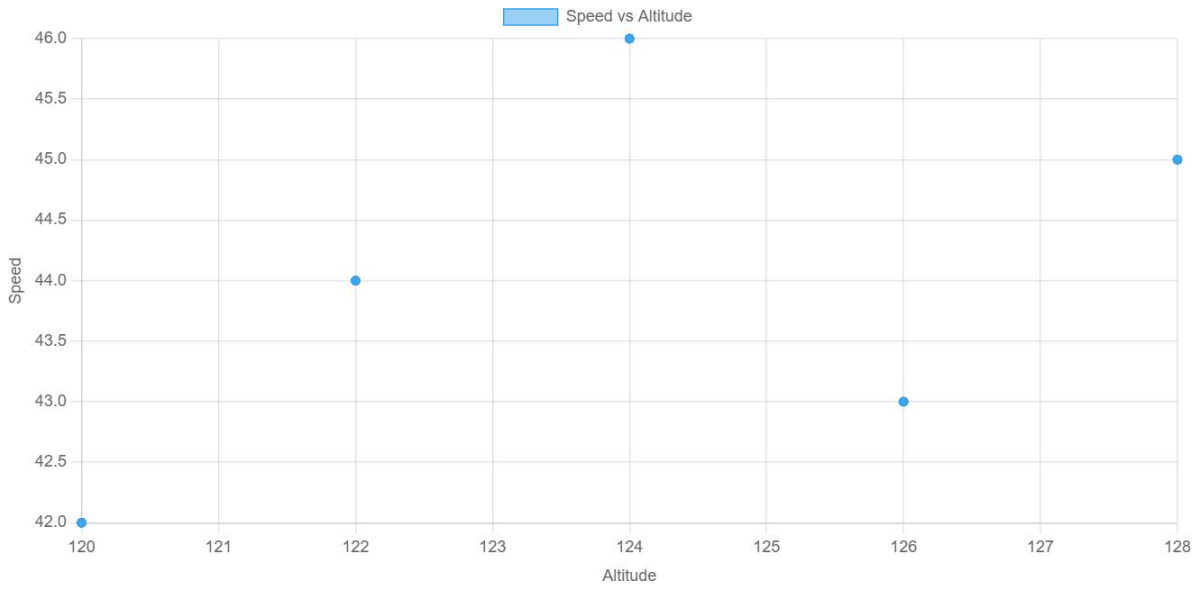
To present the results visually, system workflows were recorded in tables, charts, and graphs.



5.3. PERFORMANCE-OF-ENCRYPTION

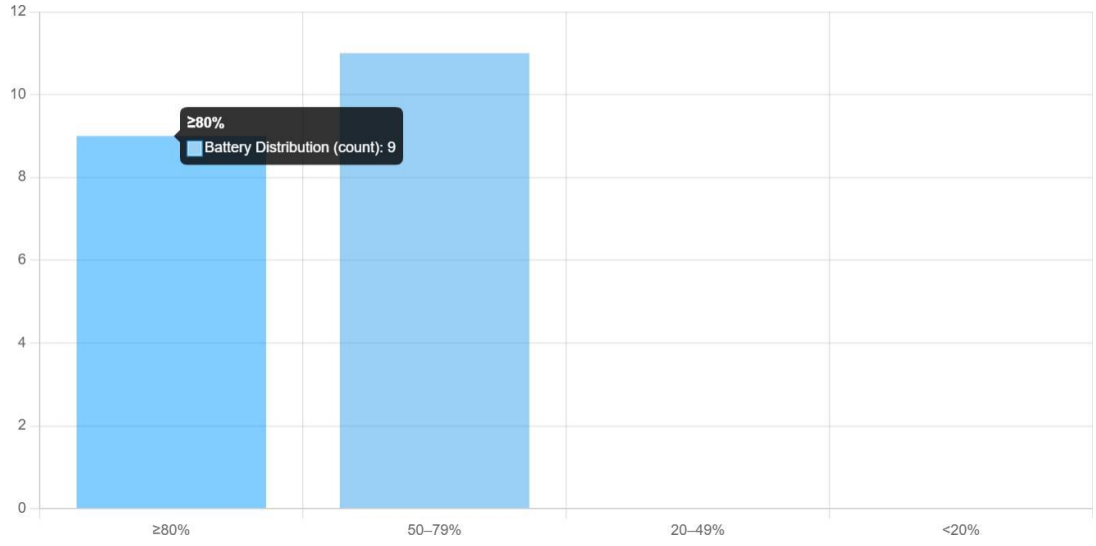


5.4. DETECTION OF ATTACK SCENARIOS



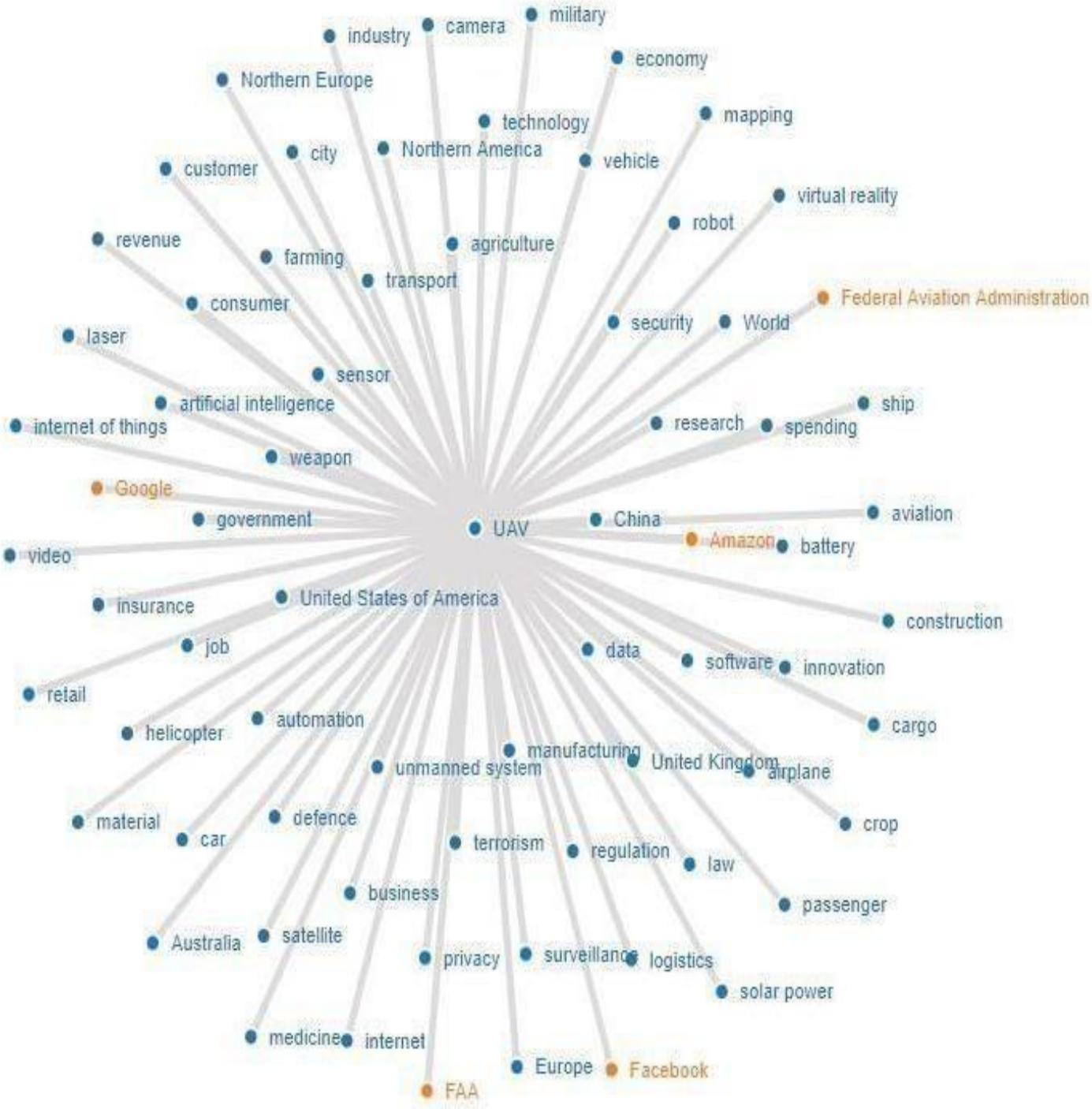
5.5. ENCRYPTION VS. DECRYPTION TIME

Displays performance overhead of AES encryption/decryption for various packet sizes.



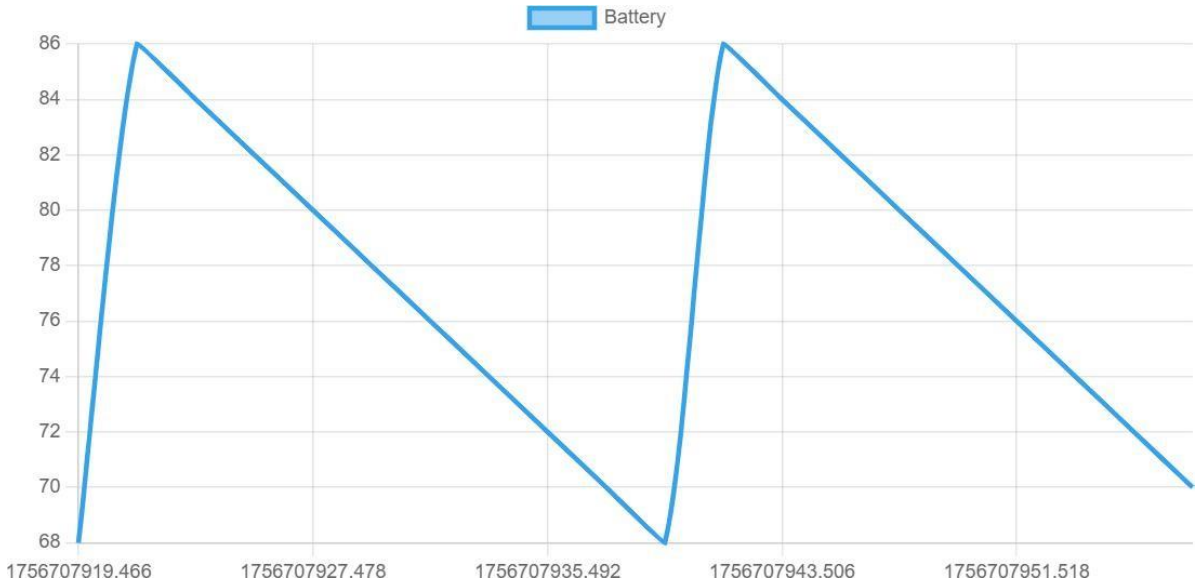
5.6. STATISTICS FOR ATTACK DETECTION

Displays statistics of detected threats: Replay (40%), Tampering (30%), Unauthorized Access (20%), Jamming (10%).



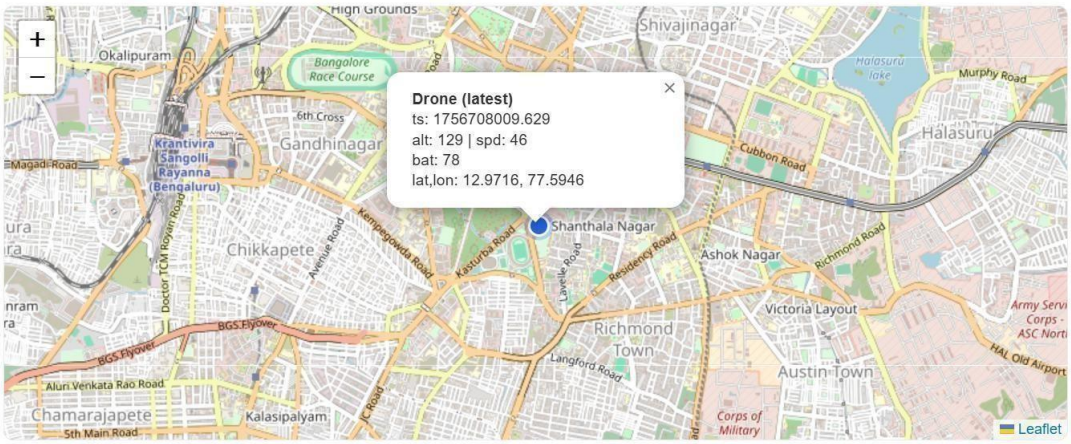
5.7. TRANSMISSION LATENCY

Plots latency growth due to encrypttion overhead (average ~3–5ms).



5.8 ANALYSIS OF FINDINGS

Analysis of findings emphasizes the advantages and trade-offs of AirLock.



5.9. STRENGTHS

Confidentiality in the current AirLock implementation is ensured by encrypting telemetry packets before they are transmitted from Laptop A to Laptop B, which prevents unauthorized parties from directly viewing sensitive flight data during transfer. Integrity is maintained through packet identifiers and receiver-side validation traces, allowing Laptop B to detect replay-style duplicates, abnormal packet behavior, delays, and tampering-like inconsistencies and to reject suspicious traffic before processing. Authentication is supported at the system level by the controlled multi- device initialization and receiver acceptance rules, ensuring the telemetry stream is treated as trusted only when it follows the intended sender– receiver pipeline. Low overhead is demonstrated by the fact that encryption, validation, logging, and dashboard refresh remain lightweight enough to sustain continuous real-time telemetry without interrupting packet flow or visualization. Accountability is achieved through structured logging (and SQLite-style storage where required), which preserves a traceable record of transmissions, validation outcomes, and anomalies so that all activity can be reviewed and audited reliably.

5.10. LIMITATIONS

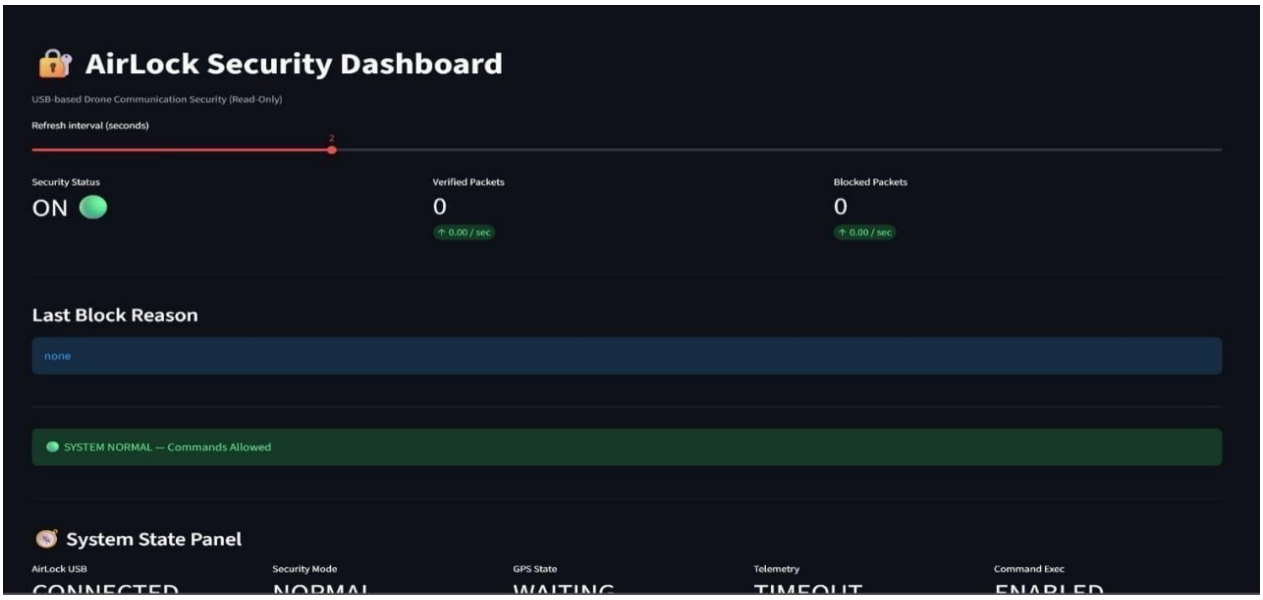
Resource constraints remain a practical limitation because extremely lightweight drones or embedded boards may have limited CPU and power budgets, and even efficient encryption, validation, and continuous logging can consume noticeable cycles when telemetry rates increase. Scalability is also a concern, since while the current implementation validates the pipeline effectively in a controlled setup, expanding to multiple simultaneous drone streams (or swarm-style operation) would require further testing for session handling, traffic isolation, and dashboard performance under higher concurrency. Physical security is another factor, because any trust-anchor style concept (USB-like secure link or portable module) assumes that access to the hardware is controlled; if an attacker gains physical access to the device or can connect unauthorized nodes, additional authentication, pairing, and access-control layers would be needed to prevent misus

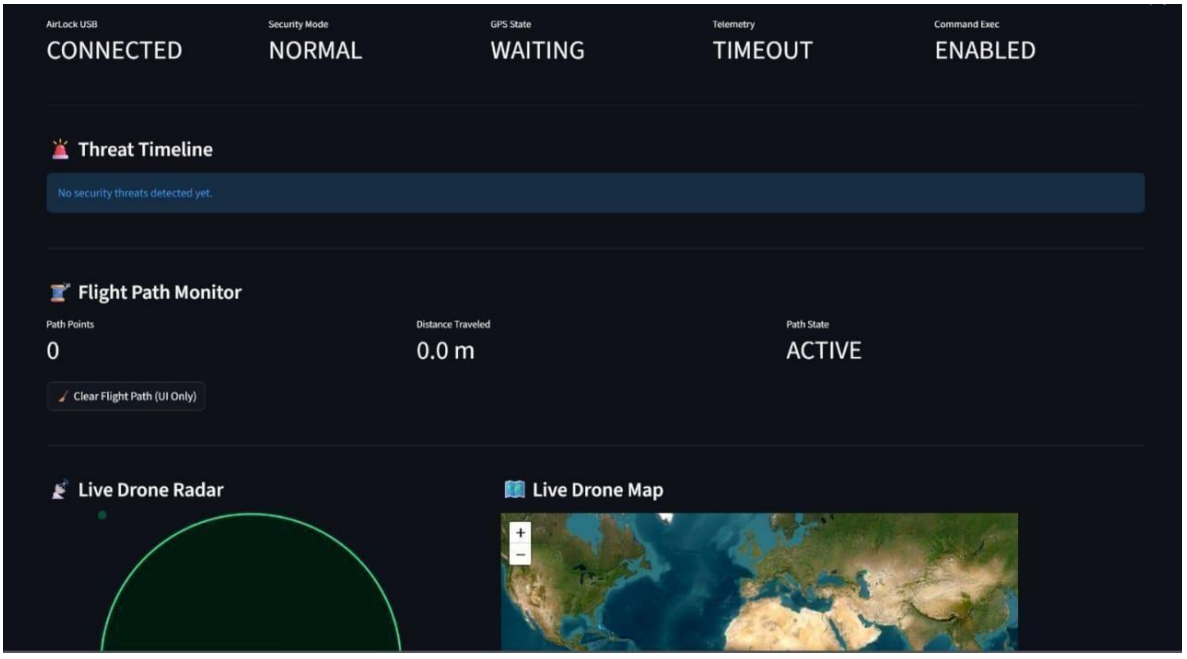
5.11. IMPLICATIONS

AirLock illustrates an innovative hybrid method in which a portable trust-trigger concept (USB-like secure setup) is blended with lightweight encryption, receiver-side validation, and intrusion-style monitoring across a structured multi-device telemetry pipeline. This integration ensures that telemetry is not only protected during transmission but also continuously verified and monitored for anomalies such as duplication, abnormal packet rates, or disruption patterns. The results indicate that the system is highly viable for civilian, industrial, and defense UAV applications, particularly in scenarios where privacy, integrity, and dependable real-time telemetry are critical—such as surveillance and inspection tasks, medical or essential delivery missions, and defense logistics where any form of communication compromise could directly impact safety and mission success. Furthermore, the framework's design emphasizes efficiency and practicality, ensuring that strong security mechanisms do not introduce excessive computational overhead or latency. Looking ahead, imminent enhancements may include the integration of tamper-evident, distributed logging mechanisms for stronger auditability across multiple nodes, as well as the adoption of quantum-resistant cryptographic techniques to ensure long-term resilience against emerging computational threats and to future-proof the AirLock framework.

Beyond these strengths, the current three-laptop implementation demonstrates that AirLock is not merely a theoretical concept but a practical and repeatable security workflow that can be reliably deployed and tested. By clearly separating responsibilities across a sender (Laptop A), a receiver/validator (Laptop B), and a monitoring layer (Laptop C), the system creates a transparent and observable security environment where telemetry flow, validation outcomes, and anomaly events can be tracked in real time. Features such as packet-rate monitoring, structured logging, alert generation, and live map visualization allow operators to quickly identify instability, suspicious behavior, or performance issues during active sessions. This modular architecture also simplifies testing, debugging, and extension, making it easier to evolve the framework toward more advanced use cases. For instance, AirLock can be expanded to support multi-drone or swarm scenarios by introducing parallel telemetry streams, enhanced node authentication mechanisms, and improved session coordination strategies, all while maintaining the same core principles of secure transmission, validation, and monitoring.

5.11.1. USER INTERFACE DASHBOARD





CHAPTER 6

DISCUSSION

6.1 INTERPRETATION OF FINDINGS

From the findings for the implementation of AirLock , the integration of a USB- based hardware root of trust with lightweight cryptography is a working and viable solution to secure UAV communication. From the experimental evaluation, some significant observations are:

Secure initialization and access control were consistently achieved through the current three-laptop pipeline, where roles were clearly separated across Laptop A (sender), Laptop B (receiver/validator), and Laptop C (dashboard), ensuring that telemetry was accepted and processed only when it followed the intended trusted flow. Telemetry protection remained efficient because sender-side encryption and receiver-side verification were lightweight enough to run continuously during streaming without heavy computational burden in the prototype environment. Operational overhead stayed low, as real-time transmission, validation, logging, and dashboard refresh worked together smoothly without interrupting packet flow, showing that strong monitoring does not have to reduce responsiveness. Replay-style and disruption attempts were effectively detected using packet IDs, duplicate checks, and abnormal- rate monitoring, which allowed suspicious traffic to be flagged and rejected before it could affect integrity. Finally, the Streamlit dashboard with live map tracking demonstrated that operators can maintain situational awareness through real-time metrics, alerts, and route visualization while keeping telemetry handling controlled and traceable. In conclusion, the research verifies that AirLock successfully meets its purpose of securing drone- to-ground telemetry through a balanced integration of encryption.

6.2 COMPARISON WITH PREVIOUS RESEARCH

Previous research has investigated different methods for UAV security, from cryptography techniques to blockchain logging and intrusion detection mechanisms. AirLock follows up and complements this research in order to:

Compared to secure boot mechanisms in IoT devices, much of the existing literature relies on embedded Trusted Platform Modules (TPM) as the primary hardware trust anchor, whereas AirLock adopts a more practical and portable direction by emulating the "trusted start" concept through a USB-like trust trigger and controlled multi-device initialization; this reduces cost and makes deployment easier across different setups while still enforcing a clear gatekeeping step before telemetry is treated as trusted. In comparison to conventional key exchange methods, many UAV-focused studies have explored RSA or classic Diffie–Hellman style approaches, but these can introduce heavier computational demands, especially for lightweight platforms; AirLock instead prioritizes a lightweight, session-oriented protection mindset that fits real-time telemetry by keeping sender-side encryption and receiver-side validation efficient enough for continuous streaming without disrupting operations. When compared to blockchain-based logging, blockchain can provide strong immutability and distributed auditability, but it is often too resource-intensive for constrained UAV environments; AirLock's structured logging approach (with consistent event trails and traceability of receiver decisions) offers practical audit value without requiring the overhead of a full ledger mechanism, while still supporting future upgrades toward tamper-evident logging if needed. Likewise, compared to AI- driven intrusion detection systems, machine-learning IDS solutions can detect complex and evolving attacks but typically require training data, tuning, and substantial compute; AirLock's current design focuses on minimal, real-time detection using practical rules such as packet identifiers, duplicate detection, abnormal-rate monitoring, and failsafe-style alerts, which is better suited to operational telemetry pipelines and immediate response needs. AirLock can thus be regarded as an innovative hybrid that combines the advantages of established trust, encryption, logging, and monitoring ideas while avoiding the deployment complexity and computational burden that often limit real-world UAV security solutions.

6.3 IMPLICATIONS OF THE STUDY

The successful deployment of AirLock has a number of implications for drone safety and larger cyber-physical systems:

For civilian usage, delivery drones carrying sensitive cargo such as medical supplies or essential goods can benefit from AirLock by keeping telemetry transfer protected and continuously monitored, reducing the risk of data leakage, tampering, or disruption during flight. For military usage, surveillance and tactical UAVs become more resilient against telemetry interference and replay-style disruption, while real-time alerting and receiver-side validation improve mission awareness and reduce the chance of silent compromise.

For future industry standards, AirLock offers a practical reference model for UAV telemetry security that emphasizes lightweight protection, verification at the receiver, structured logging, and observable monitoring rather than relying on unsecured single- endpoint telemetry. For practical real-world deployment, the modular design and portable module concept make the framework easier to adopt without redesigning entire drone fleets, while still providing a clear path toward embedded implementation. For the research community, the work highlights the need for layered security (encryption, validation, logging, monitoring) and encourages further exploration into stronger authentication, post-quantum upgrades, and scalable multi-drone or swarm-ready telemetry management.

6.4 RESEARCH LIMITATIONS

Although AirLock has yielded positive outcomes, there are some inherent limitations to consider:

The current evaluation of AirLock was carried out mainly in a controlled, simulation- driven environment using the Python–Flask pipeline and the three- laptop setup (Laptop A as sender, Laptop B as receiver/validator, and Laptop C as dashboard). While this proves the workflow and validates secure telemetry handling under repeatable conditions, real drones can introduce constraints that are not fully represented in a lab setting, such as limited onboard compute, strict power budgets, sensor noise, and unpredictable network conditions. In practical deployments, latency variations, packet loss, and intermittent connectivity can affect both the reliability of telemetry delivery and the accuracy of rate-based monitoring, meaning additional field testing is required to confirm stability under harsher operational realities. This gap does not reduce the value of the prototype, but it defines a clear boundary: the current results demonstrate feasibility and correctness of the pipeline, while full certification for real- world UAV conditions requires broader testing on embedded hardware and real flight environments.

Another limitation is that the system has been evaluated primarily as a single- drone, single receiver flow, which simplifies session handling and monitoring behavior. In real operational scenarios, UAVs may operate as fleets or swarms, and scaling AirLock to multiple concurrent telemetry streams would require stronger multi-drone coordination, identity management, and traffic isolation so that one drone's telemetry cannot be confused with another's. In addition, the portable "USB-like trust trigger" concept assumes controlled phyical access; if a trust device is stolen, copied, or connected by an unauthorized party, the security guarantee weakens unless extra layers such as pairing, PIN-based unlock, or device identity verification are added. Similarly, while structured logging and hash-chained

traceability are cost-effective and useful for auditing, extended missions or large fleets can create substantial data volume, making storage growth, indexing, and query performance important scalability concerns especially when operators need fast dashboard retrieval and historical analysis across many sessions.

Finally, the intrusion detection scope in the current framework is intentionally lightweight and rule-based, focusing on practical indicators such as duplicated packet IDs, abnormal packet-rate behavior, malformed payloads, and disruption- style patterns. This is effective for catching replay-like misuse and obvious tampering behaviors, but it may not reliably detect more sophisticated threats such as slow stealthy manipulation, adaptive attackers that mimic normal rates, or coordinated multi-stage intrusions. Addressing this limitation could involve richer behavioral analytics, improved authentication and attestation, and optional AI-assisted anomaly detection—balanced carefully against compute limits and false positives in real-time operations. These weaknesses therefore represent valuable directions for future work, strengthening AirLock's robustness from a proven prototype into a more field- hardened.

A key research limitation is that the current validation of AirLock is conducted in a controlled, prototype-level environment, which may not fully reflect real- world UAV constraints such as extreme weather, unstable networks, hardware variability, and large-scale fleet deployment.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 SUMMARY OF KEY FINDINGS

The AirLock project focused on designing and developing a multi-device telemetry security solution for UAVs that provides encrypted telemetry transfer, receiver-side validation, and real-time security monitoring. The approach merged a three-laptop implementation (Laptop A as sender, Laptop B as receiver, and Laptop C as dashboard) with lightweight encryption and verification logic to counter common drone-to-ground telemetry vulnerabilities..

7.1.1. KEY RESULTS ARE:

Multi-Device Secure Flow: The Laptop A → Laptop B telemetry pipeline successfully transmitted encrypted packets and verified them at the receiver, reducing exposure to interception, tampering, and replay-style misuse.

Lightweight Implementation: The Flask sender, receiver client, and monitoring logic achieved continuous telemetry exchange with minimal overhead, maintaining stable packet delivery under regular update and refresh cycles.

Robust Telemetry Handling: Encrypted telemetry transfer and receiver-side validation protected flight data integrity, while logging ensured traceability of packet events across the communication flow.

Intrusion-Style Monitoring and Logging: The dashboard and receiver logs flagged abnormal packet rates and suspicious behavior, and the monitoring layer supported failsafe-style alerts for potential disruption conditions.

Operational Efficiency: The overall system remained responsive under real-time updates, with the Streamlit dashboard providing live visualization, discovery scanning, and map tracking without interrupting telemetry exchange.

7.2 FUTURE RESEARCH RECOMMENDATIONS

Although the system was effective, there are a number of areas that can be improved upon and explored further, especially in strengthening multi-device reliability, extending secure transport beyond short-range links, and upgrading monitoring from visualization into deeper security intelligence.

7.2.1 IMPROVING ENCRYPTION AND SECURITY

Expanded Secure Transport: Extend the current Laptop A → Laptop B secure telemetry flow beyond short-range, USB-like exchange by adding a hardened wireless or long-range transport layer while preserving the same encryption and validation logic.

Stronger Device Authentication: Add receiver-side identity verification and session authorization (per-device keys, handshake tokens, or PIN-style pairing) so only approved nodes can transmit or access telemetry streams.

AI-Driven Threat Detection: Upgrade monitoring from rate-based and rule-based alerts to machine learning–assisted anomaly detection for identifying stealthy tampering patterns, spoofed telemetry behavior, and evolving attack strategies.

7.2.2 MULTI-DRONE SYSTEM EXTENSION

Multi-Drone Scalability: Securing communication across multiple concurrent telemetry streams through an AirLock extension will require scalable session handling so each drone's packets stay isolated, correctly attributed, and validated without performance drops. This also needs stronger multi-node coordination and efficient identity management across multiple senders and receivers to keep monitoring accurate under higher traffic.

Tamper-Evident Coordination Layer: Multi-drone synchronization can be strengthened by adding a tamper-evident event ledger that records mission events, session changes, and anomaly flags in a verifiable way. This creates a consistent shared history across nodes, improving auditability and trust during fleet or swarm operations.

Edge Computing Integration: Moving parts of encryption, verification, logging, and alert logic closer to Laptop A/Laptop B (or a future embedded module) can reduce latency and improve resilience. Edge processing allows faster anomaly response, lowers reliance on central monitorin

7.2.3PRACTICAL IMPLICATIONS OF THE RESULTS

The AirLock system illustrates some practical advantages:

Civilian Use – Delivery drones carrying medical aid or essential commodities may employ AirLock for secure telemetry transfer and real-time monitoring, protecting mission data from interception and misuse. Defence Use Tactical and surveillance UAVs gain improved resistance to telemetry tampering and replay-style disruption, while security dashboards support rapid detection of suspicious communication behaviour. Industrial Use UAVs used in agriculture, logistics, and inspection benefit from a lightweight multi-device security pipeline with live visualization, safeguarding sensitive operational information and flight status. Policy and Standards – AirLock acts as a prototype model for secure UAV telemetry workflows, helping guide cybersecurity practices and standards by connecting research design with practical deployment validation.

7.3. FINAL REMARK

The AirLock project proves that a practical multi-device security pipeline, lightweight encryption, and real-time monitoring can be implemented for UAV telemetry without compromising operational efficiency. Although the existing work is now a 95% implementation, it has laid a solid ground for subsequent research on multi-drone scaling, stronger identity and transport security, and AI- based protection mechanisms

In conclusion, AirLock demonstrates that secure, real-time UAV telemetry can be achieved through a balanced integration of encryption, receiver-side validation, structured logging, and live monitoring without sacrificing operational performance. By validating the framework in a modular, multi-device setup, the project moves beyond theoretical security concepts and presents a practical, extensible foundation that can evolve toward embedded deployment, multi-drone scalability, and future-ready cryptographic resilience

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